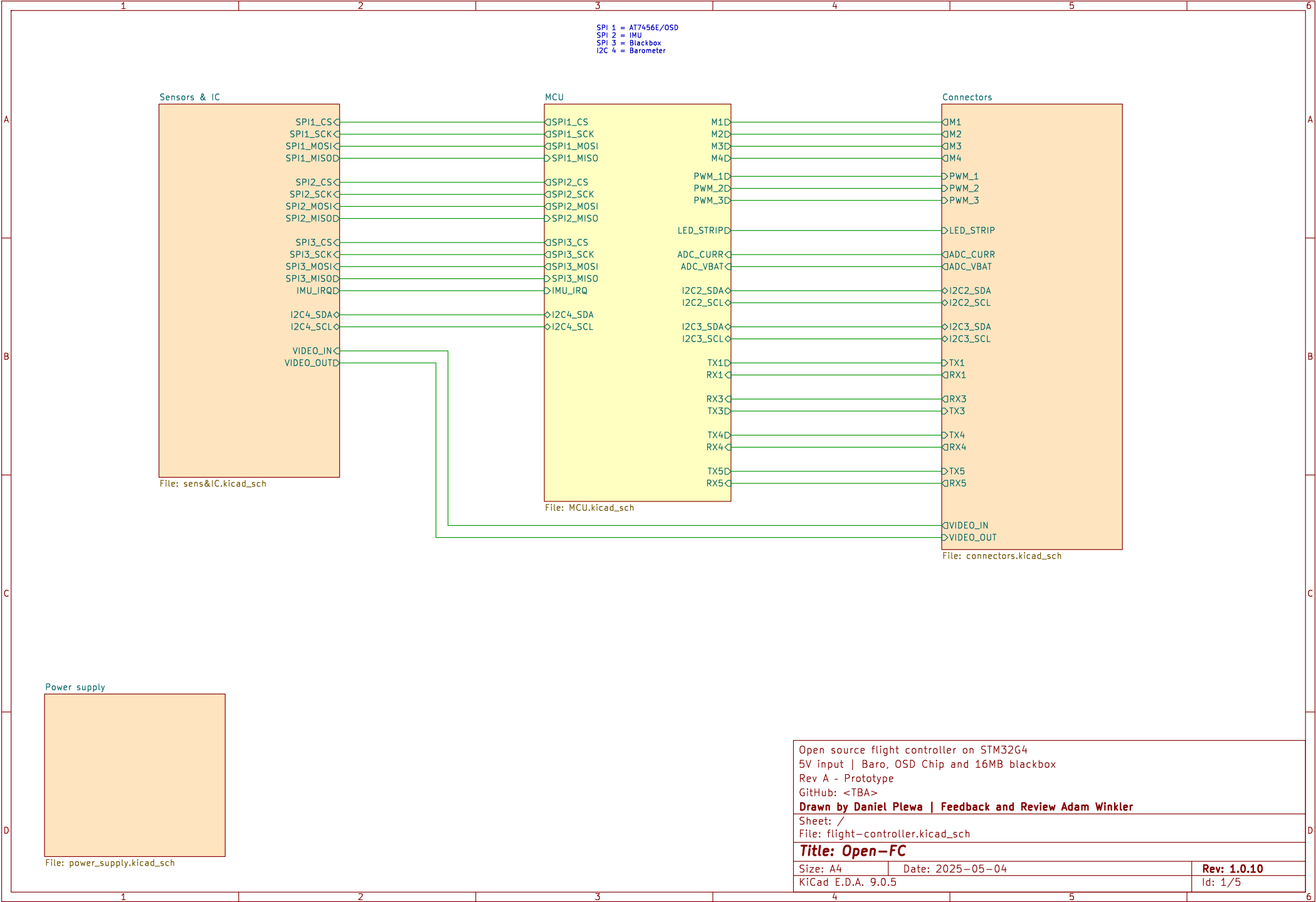
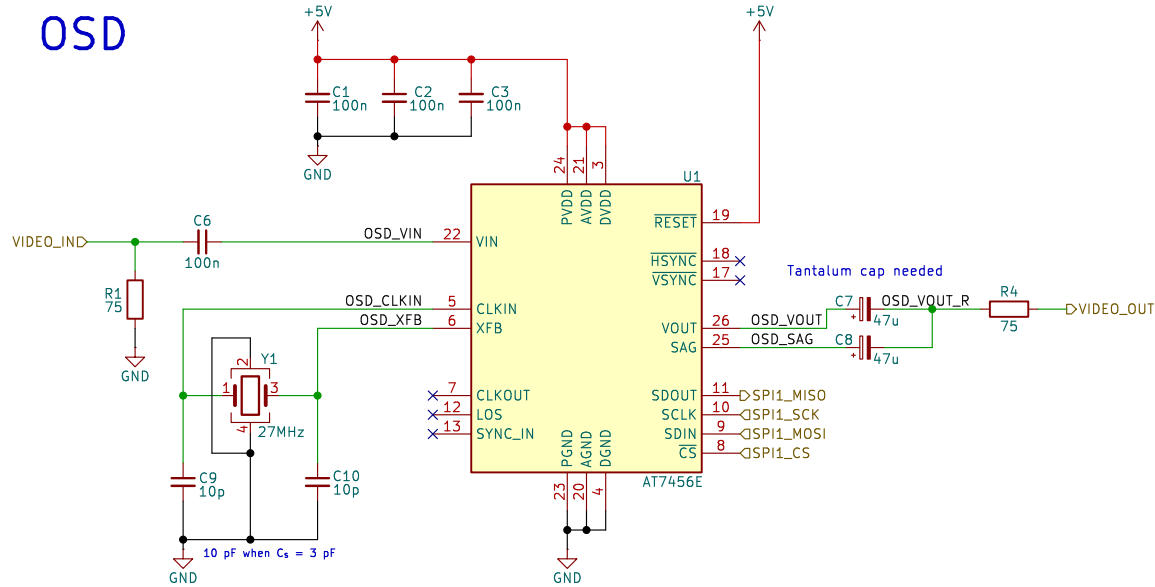
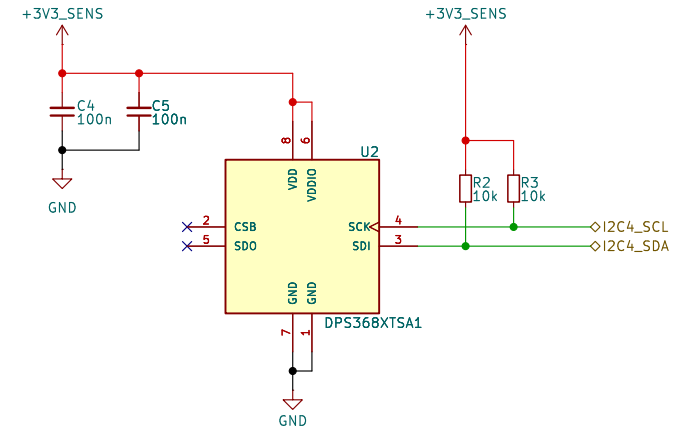




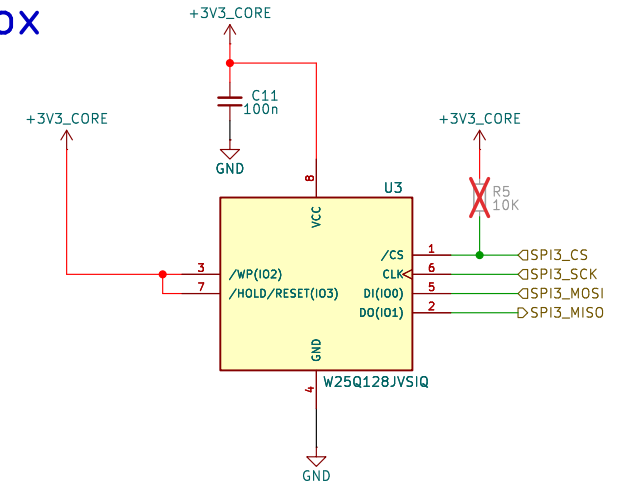
OSD



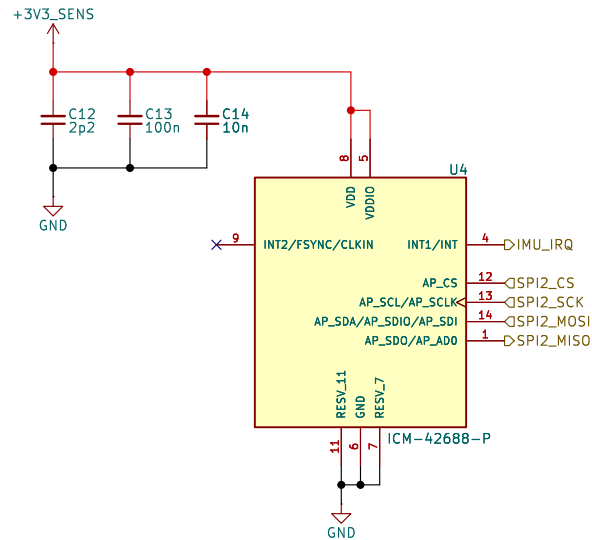
Baro



Blackbox



IMU



Open source flight controller on STM32G4
5V input | Baro, OSD Chip and 16MB blackbox
Rev A - Prototype
GitHub: <TBA>

Drawn by Daniel Plewa | Feedback and Review Adam Winkler

Sheet: /Sensors & IC/
File: sens&IC.kicad_sch

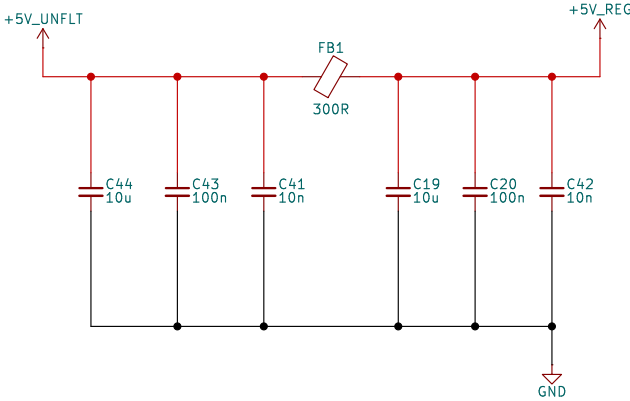
Title: AT7456E, IMU, Blackbox, Barometer

Size: A4 | Date: 2025-05-04
KiCad E.D.A. 9.0.5

Rev: 1.0.10
Id: 2/5

Buck converter
5V 3A

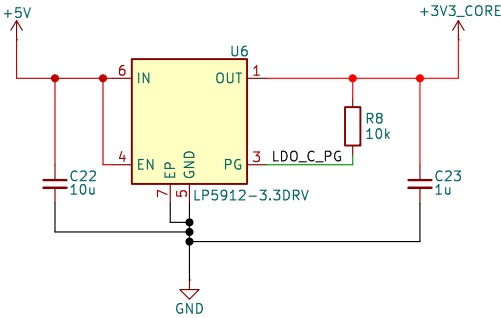
Power Path
5V_UNFLT -> 5V_REG -> 5V -> 3V3_CORE
USB -> VBUS -> 3V3_SENS



Main power source

Design and CAP size
<https://webench.ti.com/appinfo/webench/scripts/SDP.cgi?ID=45E85F2F9E19CD80>

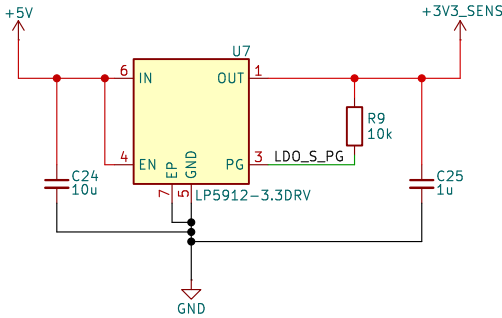
LDO
3V3 500mA



LDO 3.3V for MCU and Blackbox

Design
<https://webench.ti.com/appinfo/webench/scripts/SDP.cgi?ID=F7ED04D030C13C86>

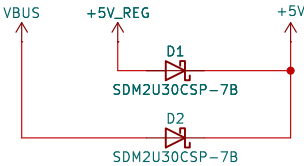
LDO
3V3 500mA



LDO 3.3V for IMU and Baro

Design
<https://webench.ti.com/appinfo/webench/scripts/SDP.cgi?ID=F7ED04D030C13C86>

Power switch



Open source flight controller on STM32G4
5V input | Baro, OSD Chip and 16MB blackbox
Rev A - Prototype
GitHub: <TBA>
Drawn by Daniel Plewa | Feedback and Review Adam Winkler

Sheet: /Power supply/
File: power_supply.kicad_sch

Title: Power supply, LDO

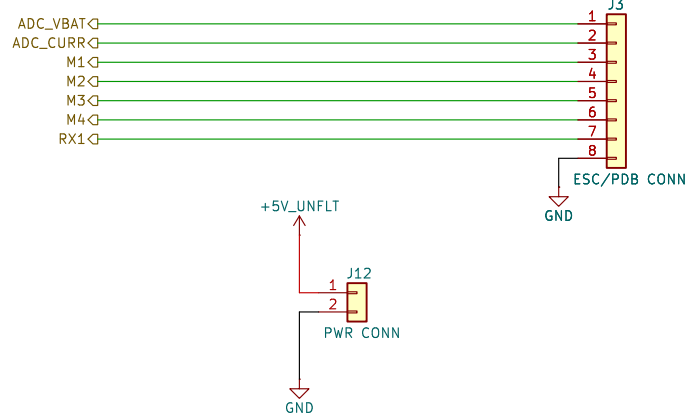
Size: A4 Date: 2025-05-04

KiCad E.D.A. 9.0.5

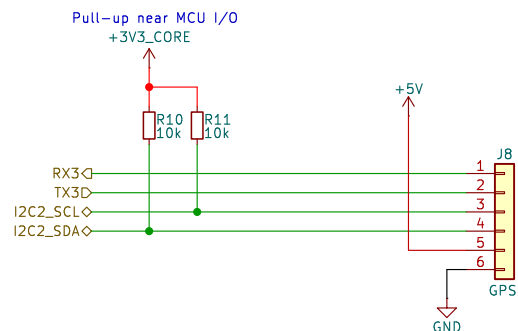
Rev: 1.0.10

Id: 3/5

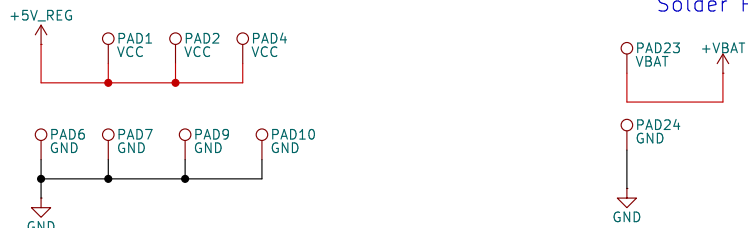
ESC/PDB CONN PWR CONN



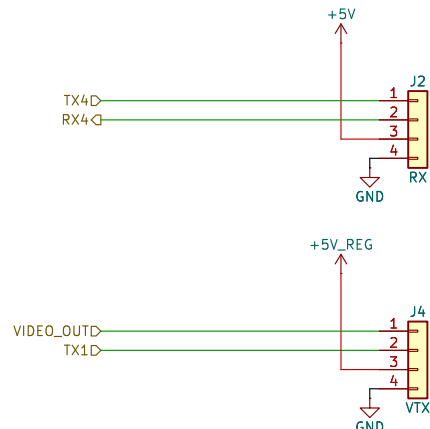
GPS CONN



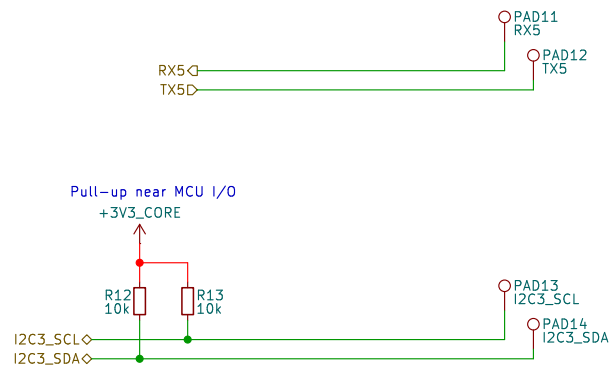
Solder Pad



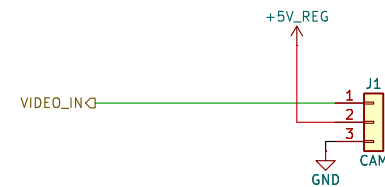
RX and VTX CONN



Additional BUSES



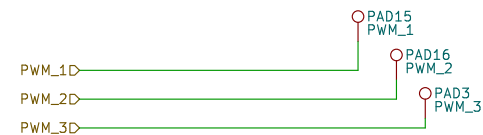
CAM CONN



LED CONN



Additional PWM OUT



Pin assignment on the ESD protection IC can be rearranged for layout optimization.
Feel free to swap ESD IC pins if it helps with layout.
Check total power consumption – the JST-PH connector has limit of 3A.
Preferred max current consumption is 2.5A

Open source flight controller on STM32G4
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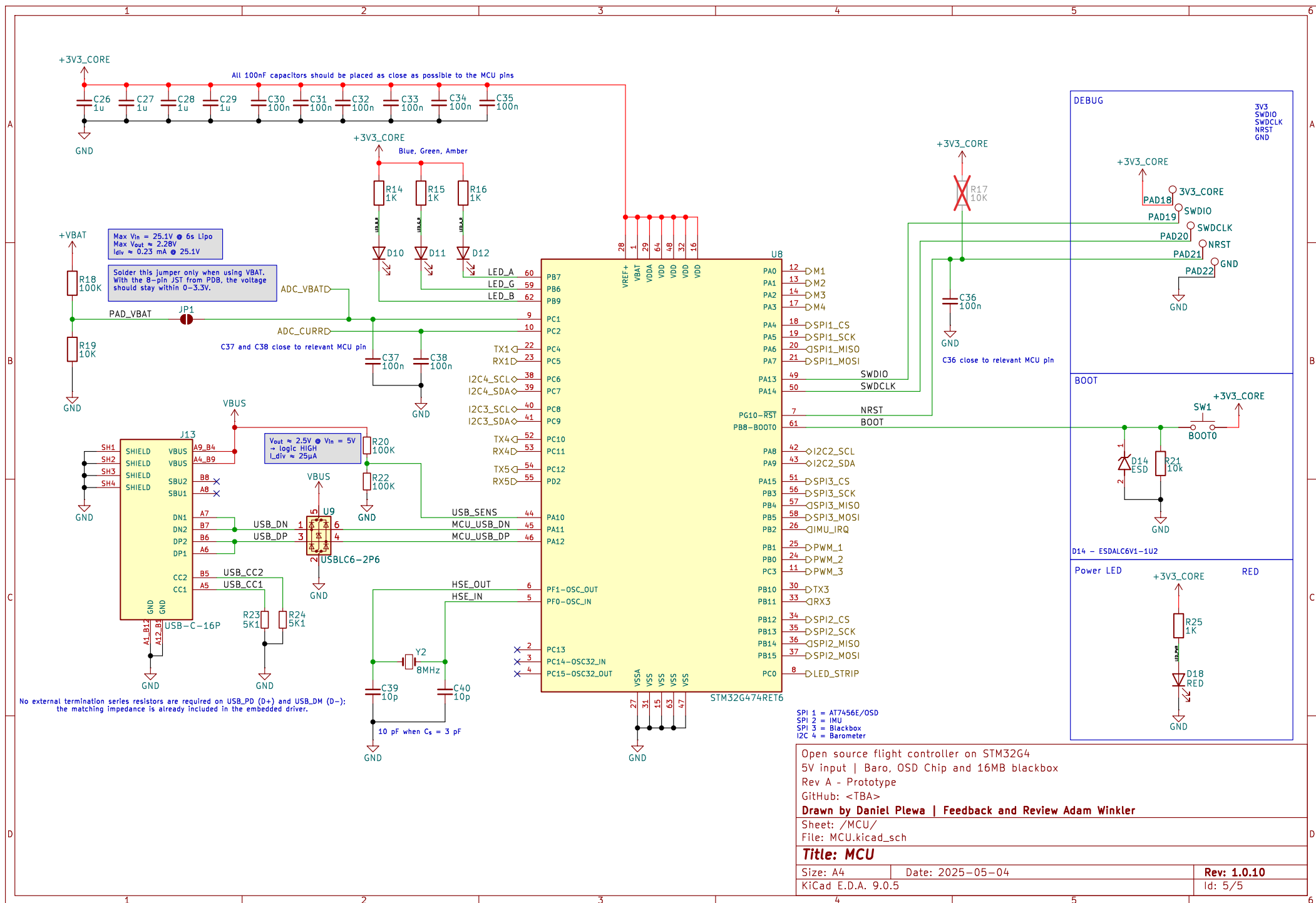
Drawn by Daniel Plewa | Feedback and Review Adam Winkler

Sheet: /Connectors/
File: connectors.kicad_sch

Title: Connectors

Size: A4 | Date: 2025-05-04
KiCad E.D.A. 9.0.5

Rev: 1.0.10
Id: 4/5



SPI 1 = AT7456E/OSD
 SPI 2 = IMU
 SPI 3 = Blackbox
 I2C 4 = Barometer

Open source flight controller on STM32G4
 5V input | Baro, OSD Chip and 16MB blackbox
 Rev A - Prototype
 GitHub: <TBA>
 Drawn by Daniel Plewa | Feedback and Review Adam Winkler

Sheet: /MCU/
 File: MCU.kicad_sch

Title: MCU	
Size: A4	Date: 2025-05-04
KiCad E.D.A. 9.0.5	Rev: 1.0.10
	Id: 5/5